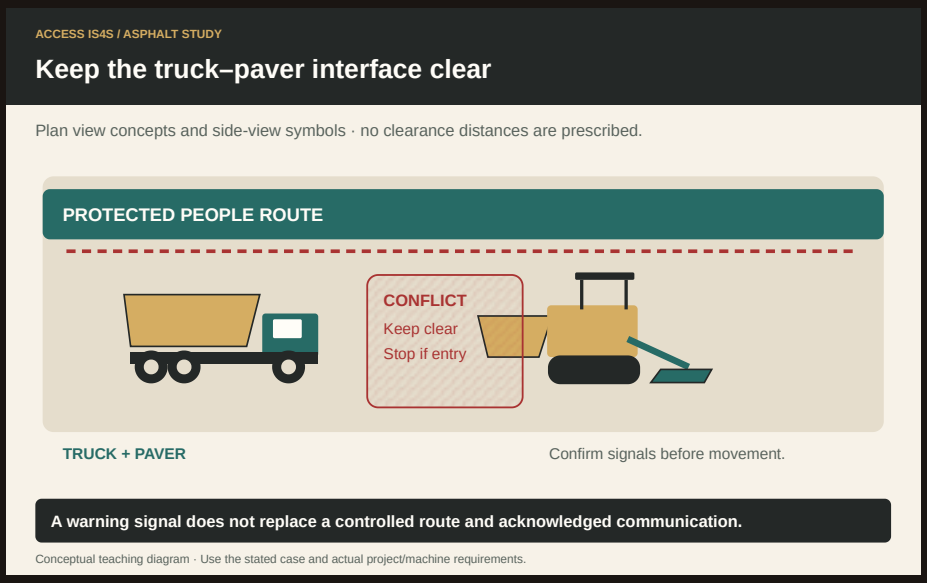
ACCESS · INDIGENOUS SKILLS 4 SUCCESS

People, tools and safe work

Volume 2 · B2, B3, B4, B5, B6, B7, B8, B9, B10

Classroom development edition · 30 October 2026



ACCESS IS4S. (2026). Keep the truck–paver interface clear [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

In this volume

1. How to use the volume and key terms
2. Working safely around haul trucks
3. Safety working around and operating power brooms
4. Safety working around and operating air hammers
5. Safety working around and operating propane tanks
6. Weather effects and precautions
7. Safety and traffic control procedures
8. Plant site and gravel pit safety
9. Light vehicle transporting procedures
10. Complete fatigue awareness training
11. Prepare a clear work-zone briefing

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for people, tools and safe work. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

Study chapter · Outline B2

Working safely around haul trucks

A haul truck brings material into a place where people and machines already have work to do. The transfer becomes a coordinated operation: truck, paver, ground worker and traffic control each need a shared understanding. **Blind area** means a space the operator cannot see from the operating position. **Pinch point** means a place where a person can be caught between moving or closing surfaces. Neither disappears because a worker is wearing high-visibility clothing.

At Cedar Learning Access, a delivery may arrive while someone needs the community entrance. The crew must protect both the delivery path and the public route. The task is to keep people out of the interaction between truck and paver while the approved delivery arrangement is followed.

Consequences of lapses in communication

Equipment specific hazards

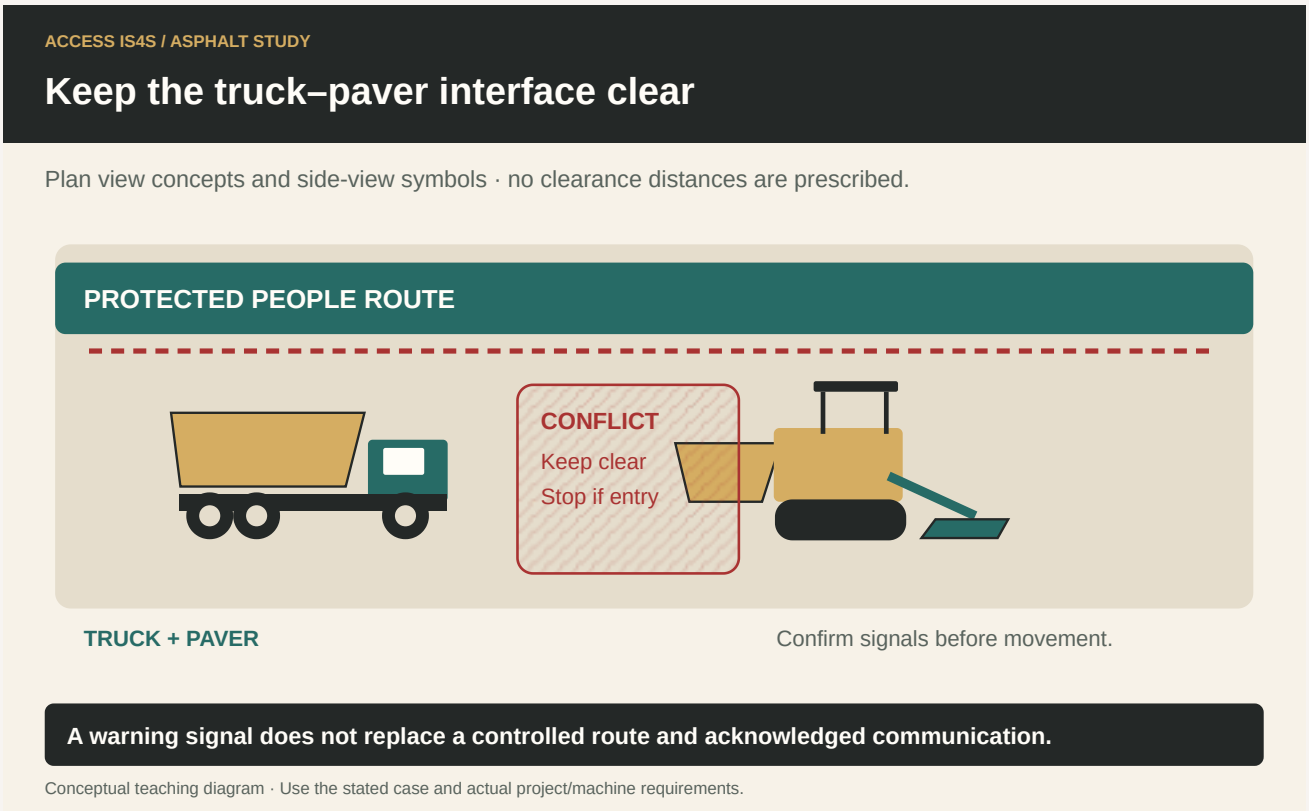
LEARNING STEP 1

Consequences of lapses in communication

Communication fails when a message is unheard, misunderstood or directed at the wrong person. Agree the signalling arrangement before the approach: the designated person, the signals or radio channel, the acknowledged stop instruction and the response to lost contact. Use the established site procedure, precise messages and confirmation. A driver looking toward a worker is not proof that the worker's intention was understood. Eye contact supports communication, but it does not replace a confirmed message or a safe position.

The ground worker identifies themselves and remains in the planned visible position with an escape route, away from the truck's swept path and reversing space. If visual or radio contact is lost, the movement must stop under the agreed procedure; the worker does not approach a moving vehicle to restore contact. Restart only after the people involved have re-established the arrangement. Record a recurring communication failure so the system can be improved.

Keep the truck–paver interface clear



Note. ACCESS IS4S. (2026). Keep the truck–paver interface clear [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

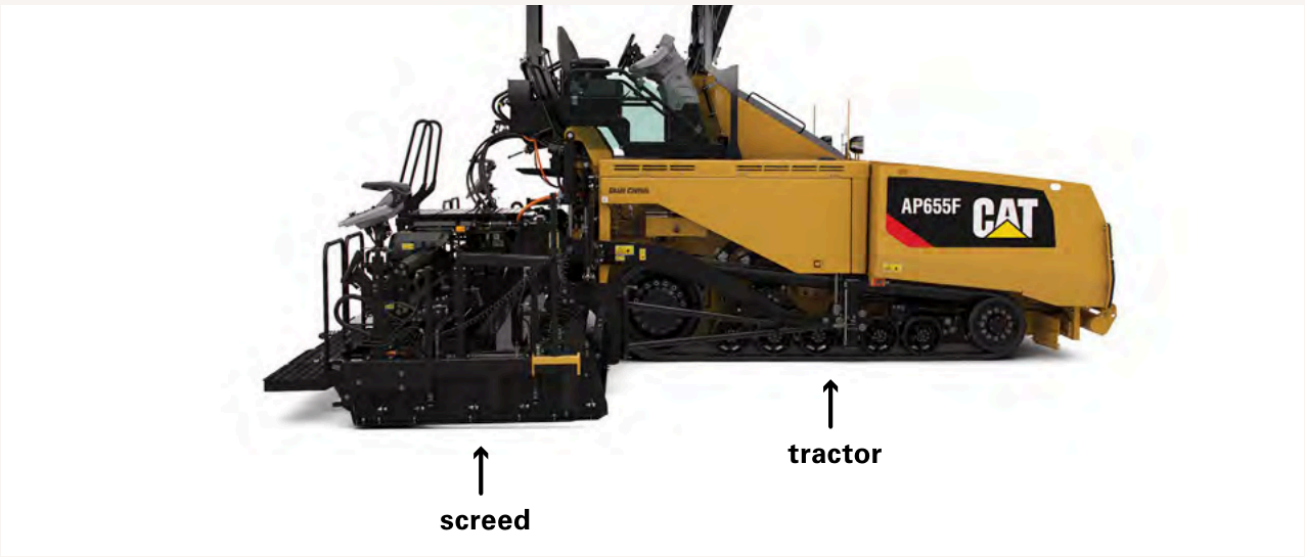
LEARNING STEP 2

Equipment specific hazards

Study the whole movement envelope, not only the tyre tracks. A turning truck has a swept path; the body, rear overhang and attachments can move through different areas. A reversing truck creates a changing space between itself and the paver. That space stays clear of people and loose equipment. A raised body changes height and stability; overhead utilities, structures and ground condition must be addressed before the delivery operation.

Clean mirrors, windows, cameras and lights help the operator see, but visibility aids have limits and require inspection. Spilled material, a dirty access step or a contaminated walking surface adds slip and housekeeping risks. A person should never infer the safe location from an illustration’s scale. The actual truck configuration, manufacturer instructions, assessment and traffic arrangement determine the controlled area. A pre-delivery check connects these physical conditions to the agreed communication.

Two main sections of the paver



Note. Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888, source PDF p. 63). Faithful crop; original labels retained. Conceptual paver example in the 2016 application guide; this picture does not identify a specific machine serial number. Full source context.

Worked example · A missed acknowledgement

Fictional sequence: 08:10 the ground worker calls the truck identifier; 08:11 an unrelated driver replies; 08:12 the assigned driver begins backing. The assigned driver has not acknowledged the direction. A pedestrian also approaches the facility entrance.

Think it through.

The sequence contains no confirmed instruction between the assigned people. Stop the movement through the agreed procedure, protect the pedestrian route and re-establish the identified driver/signaller exchange. The three timestamps show order, not proof of a safe movement. The handover should record the wrong-driver reply and the access conflict; simply repeating the original call leaves the cause unresolved.

Try it and explain your thinking

Explain and apply the idea

Annotate the truck/paver interaction diagram with blind-area, pinch-point, swept-path and overhead-hazard questions. Describe the location a ground worker must avoid and what must be confirmed before choosing a signalling position.

Skills you use: Reading, Problem solving, Communication.

Calculate and check the result

Rewrite the three radio messages in the worked case so the intended driver and acknowledgement are unambiguous. Rehearse a lost-contact event with spoken classroom roles only.

Skills you use: Writing, Communication, Collaboration.

Create and save a one-page delivery brief for the fictional site

Create and save a one-page delivery brief for the fictional site. A partner adds a blocked entrance and dirty camera. Revise the brief, reopen the saved version and identify your change.

Skills you use: Digital, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
In a controlled, authorized exercise, the instructor checks the learner's visible safe position and escape route against the actual plan.	0 / 1 / 2	Evidence/help: _____
Observe correct communication, loss-of-contact response and keeping the truck/paver gap clear; stop if the arrangement fails.	0 / 1 / 2	Evidence/help: _____
Record the machine, site conditions, supervision, observed actions and repeat attempt. Classroom role-play alone does not demonstrate vehicle-side competence.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B2, printed p. 28; source PDF pp. 30–30.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 18: Traffic Control. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline B3

Safety working around and operating power brooms

A power broom prepares a surface by moving unwanted loose material. Its brush also moves air, dust and debris. A clean-looking surface may still have an uncontrolled airborne exposure, and a clear brush path may cross a pedestrian route. Read the machine as a combination of **travel**, **brush movement** and **dust control**.

The Rosco RB50 example in this chapter is the manufacturer's Operations, Service and Parts Manual 1017258-03, Rev. B, for serial 255027 and above. Its component and warning pages are a real model study. Match a different training machine to its own manual before practical instruction.

Effects of noise and dust

Dangers associated with vehicle movement

Dangers associated with moving parts

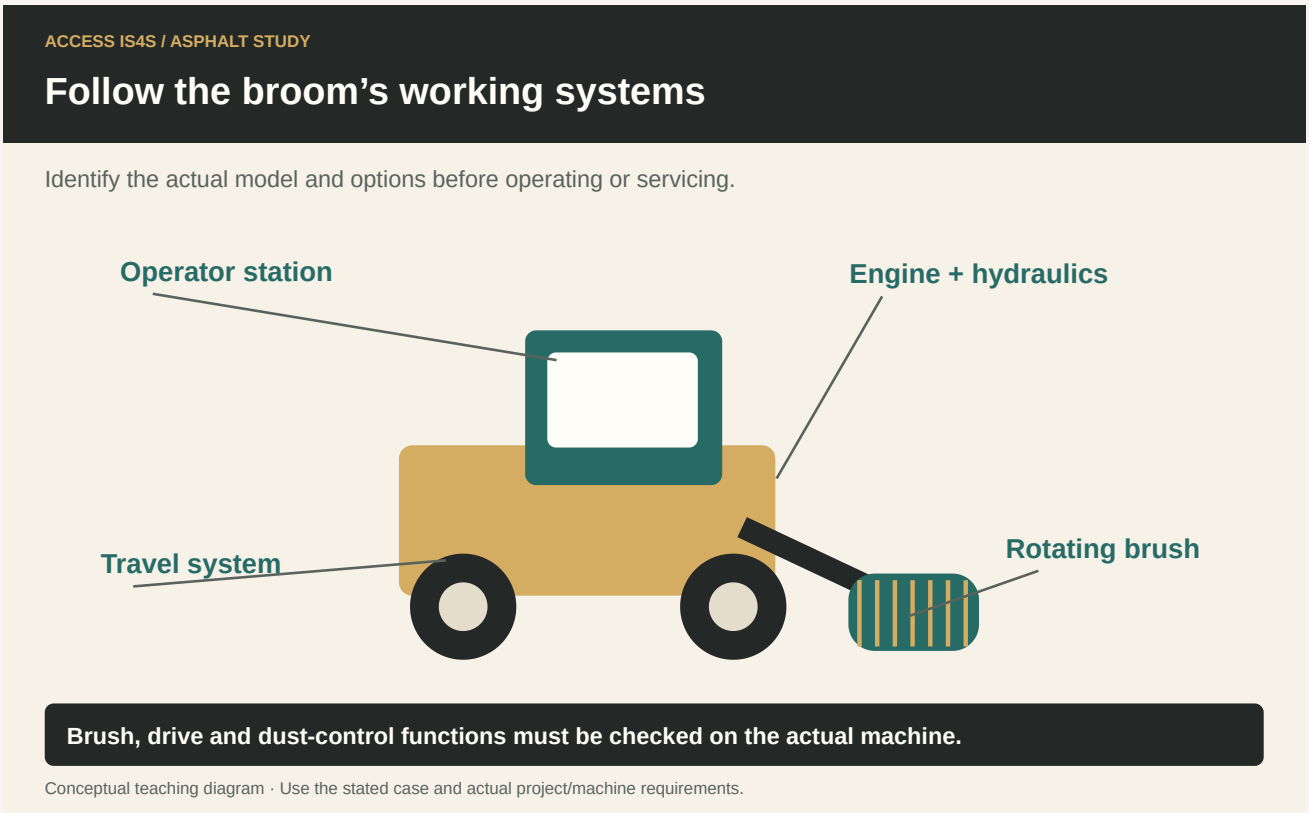
LEARNING STEP 1

Effects of noise and dust

Noise can damage hearing and mask a spoken warning. Dust can irritate or contain hazardous constituents, including respirable silica where the material produces it. Identify the material and exposure conditions; visible dust is not a measurement of the respirable fraction. Controls may include the machine's appropriate dust-suppression system, a suitable method and access separation, with exposure assessment and required respiratory/hearing programs.

PPE is chosen after the hazard assessment. High visibility addresses conspicuity; it does not control inhalation. Respiratory protection must match the contaminant, exposure and required fit/program conditions. Inspect the actual water system or other fitted dust-control equipment under the manual, and respond if its performance changes. Do not assume every RB50 has every optional system shown in a product picture.

Follow the broom’s working systems



Note. ACCESS IS4S. (2026). Follow the broom’s working systems [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Dangers associated with vehicle movement

Travel and steering can create a collision even when the brush is disengaged. Establish the working route, speed/movement controls, pedestrian separation, traffic coordination and operator visibility before sweeping. Broom swing and attachments can extend beyond the tyre path. A person at the edge of the travel lane may still be within the moving assembly’s envelope.

The operator checks access and the surroundings from the intended operating position and follows the machine’s travel and sweeping instructions. The ground crew remains in the planned safe area. A new parked vehicle, visitor or traffic queue changes the route; the operator stops and the arrangement is reassessed. A warning horn tells people something is happening but does not give the machine a clear route.

LEARNING STEP 3

Dangers associated with moving parts

The rotating brush, drive mechanism, hydraulic connections and moving linkages can entangle, strike or pinch. Guarding and exclusion protect people during use; inspection, cleaning and adjustment require the applicable shutdown and hazardous-energy control. A stopped brush may still be supported by hydraulics

or connected to stored energy.

Trace the component picture from brush core and bristles to support/linkage and machine. Identify which parts rotate, swing or raise. Loose clothing, hands or tools must not enter the moving mechanism. A jam or wound material is reported and dealt with under the actual isolation procedure. Reaching in for a quick pull while someone holds a control is not isolation.

Rosco RB50 machine overview and component key

MACHINE OVERVIEW

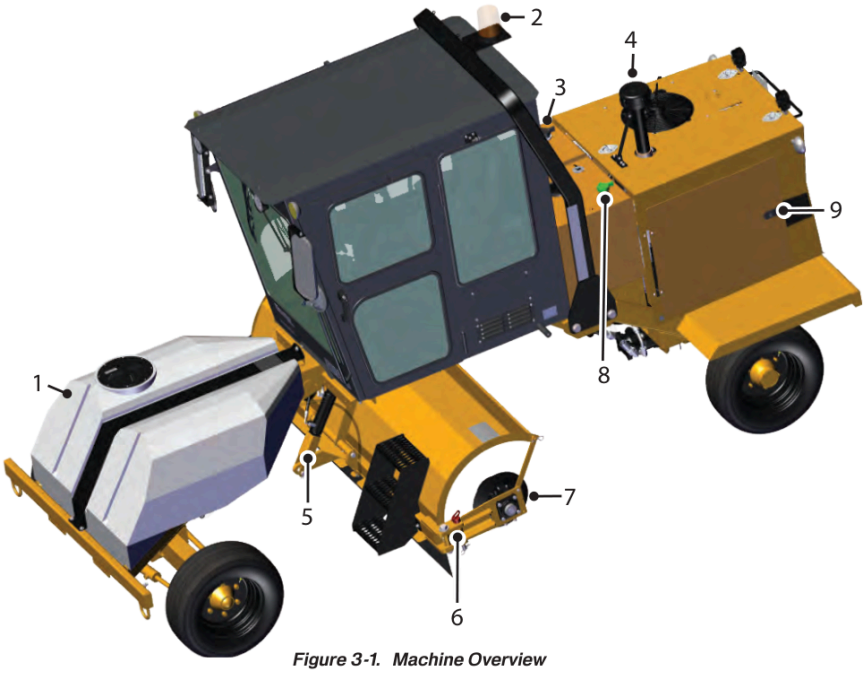


Figure 3-1. Machine Overview

ITEM NO.	COMPONENT NAME	FUNCTION
1	Water Tank (Option)	Water tank mounted on front of machine if equipped with optional Dust Suppression System. Holds 150 gallons.
2	Beacon Light	Flashing beacon light for added safety to alert traffic of broom operation. The beacon light is removable.
3	Hydraulic Oil Filler Cap	Hydraulic oil filler cap contains strainer to prevent debris from entering system when filling hydraulic tank.
4	Air Intake Stack	Rotary pre-cleaner.
5	Brush Lock Arm	Manual lock arm for locking brush into place (position shown) during travel and transport.
6	Brush Core Lock/Release	Lock pin releases and locks brush core into place when changing brush. No tools are required to change brush.
7	Brush	Angles 45 degrees right or left. 32-inch steel brush tube with many different bristle options.
8	Fuel Filler Cap	Fill using only ULSD (Ultra-Low Sulfur Diesel) fuel.
9	Engine Access Panel	Locking panel to access engine (on each side of the machine).

Note. Rosco / LeeBoy. (n.d.). RB50 Broom: Operations, service and parts manual (1017258-03, Rev. B, Fig. 3-1, p. 3-5; PDF p. 43). Serial 255027 and above. Full source context.

Worked example · The clean strip and the dusty doorway

Fictional observation: a broom produces a visibly clean 80 m strip, but dust travels toward an occupied doorway and its fitted water system has stopped spraying. The operator proposes finishing the last 20 m first.

Think it through.

The achieved 80 m does not validate the exposure control. The planned run is 100 m; 20% remains, but that arithmetic is irrelevant to accepting an uncontrolled hazard. Stop the affected work under the procedure, protect the doorway and have the dust-control fault assessed. Resume when the task controls are established, not when the inconvenience feels small.

Try it and explain your thinking

Explain and apply the idea

Identify brush, supports, operator position and any dust-control components visible in the RB50 manual image. Separate visible features from optional equipment that must be checked.

Skills you use: Reading, Communication.

Calculate and check the result

Check the worked-case percentage and write a defect/exposure report that distinguishes work completed from safe conditions.

Skills you use: Numeracy, Writing, Problem solving.

Use a device to locate the manual's component, safety and sweeping sections

Use a device to locate the manual's component, safety and sweeping sections. Save three page references, then redesign your brief when the route crosses a public entrance.

Skills you use: Digital, Adaptability, Creativity and innovation, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Match the broom, serial range and fitted options before demonstration.	0 / 1 / 2	Evidence/help: _____
Observe inspection and use of selected PPE, route awareness and safe ground-worker positioning during an authorized sweeping exercise.	0 / 1 / 2	Evidence/help: _____
Observe response to a simulated dust-control fault with the machine safely controlled; assess isolation knowledge separately from operating skills.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B3, printed p. 29; source PDF pp. 31–31.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 6: Substance Specific Requirements. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 7: Noise, Vibration, Radiation and Temperature. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 12: Tools, Machinery and Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.

- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 18: Traffic Control. Open publication.
- Rosco / LeeBoy. (n.d.). RB50 Broom: Operations, service and parts manual (1017258-03, Rev. B). Serial 255027 and above. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline B4

Safety working around and operating air hammers

An air hammer converts compressed-air energy into repeated impacts at its working tool. **Air hose**, **coupling**, **retainer**, **handle**, **trigger/control** and **tool steel** are parts to identify before learning the task. The energy that makes breaking possible can also create noise, vibration, flying material, hose movement and physical strain. Preparation therefore includes the air supply and connections as well as the hammer itself.

Noise and vibration hazards

Hazards of flying debris

Ergonomics and safe equipment handling

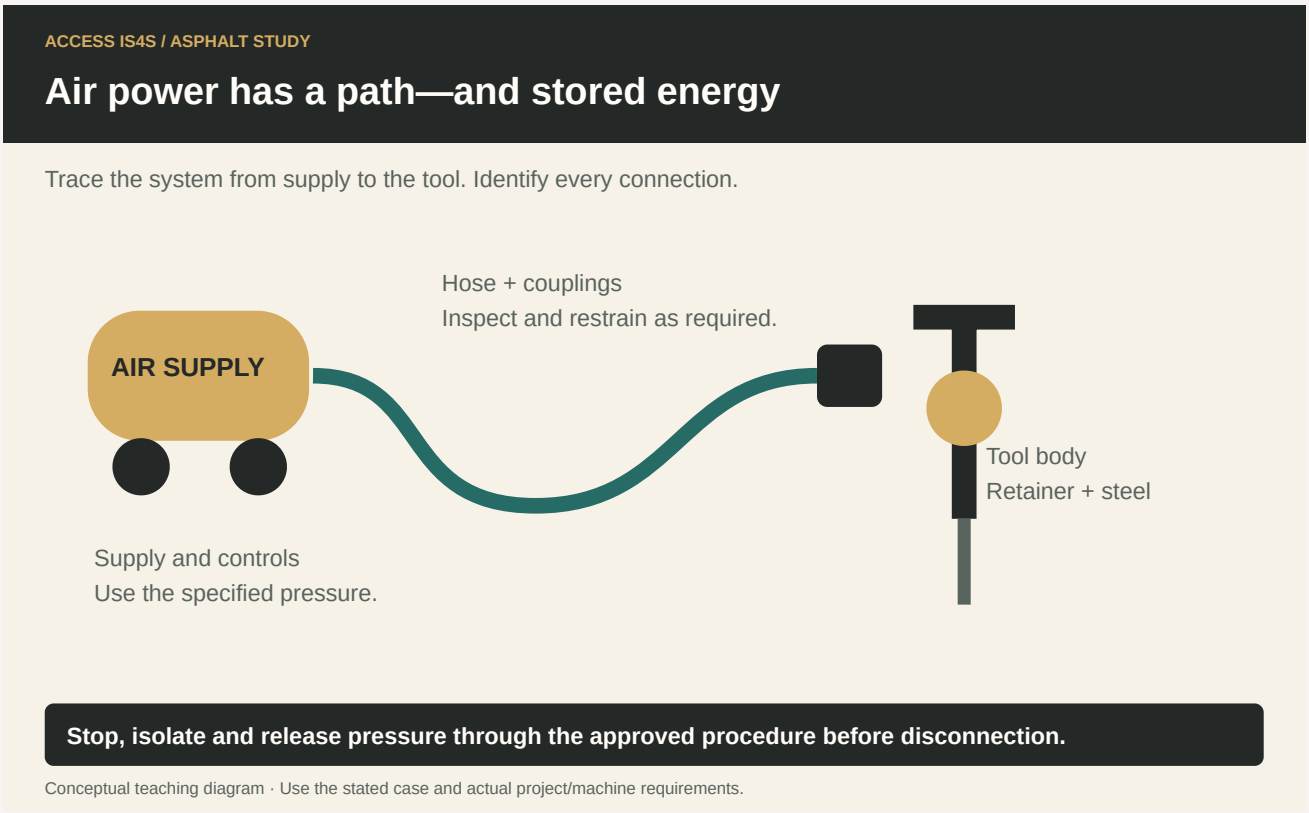
LEARNING STEP 1

Noise and vibration hazards

Repeated impact produces substantial noise and hand-arm vibration. Exposure depends on the actual tool, maintenance, material, use time and working conditions. A rating in a brochure is not an individual's measured daily exposure. Use the assessed controls: appropriate equipment, maintenance, exposure organization and correctly selected hearing/PPE arrangements. Report unusual vibration, looseness, numbness or other concerns through the workplace process.

The CP1210/1230/1240 family manual provides actual pneumatic-tool safety and component information; the applicable model and configuration still matter. A lower-vibration model may improve control but does not eliminate the need to assess the job. For a training record, distinguish trigger time from total time spent in the work area.

Air power has a path—and stored energy



Note. ACCESS IS4S. (2026). Air power has a path—and stored energy [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Hazards of flying debris

Impact breaks material into fragments that can strike the operator or others. Assess the material, access and nearby work; establish the required controlled area and suitable protection. Eye/face protection must match the hazard, while dust controls address the airborne material. Positioning a bystander just outside a sketch's boundary is insufficient without the actual assessment.

Inspect the tool retention and hose/coupling arrangement under the manual. A loose connection can become an uncontrolled hose, and an unsecured tool can be ejected. Before disconnecting or changing a tool, isolate the supply and release stored pressure using the specified procedure. Never test for a leak with a hand or direct compressed air at a person.

LEARNING STEP 3

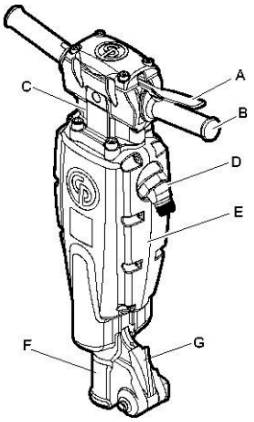
Ergonomics and safe equipment handling

A heavy tool can strain the body while carrying, starting, working or stowing. Plan access and the working position before lifting it into place. Choose the appropriate tool/support and approved handling method; avoid twisting with the load or using the body to force an unsuitable tool. Hoses should be routed to reduce tripping, dragging and damage while allowing the intended controlled movement.

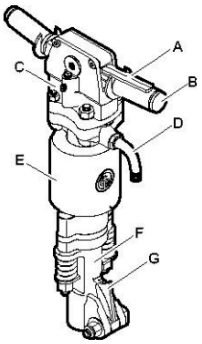
Maintenance affects effort: damaged or unsuitable tool steel can increase the force and time a person applies. The operator recognizes a change and reports it instead of compensating with more force. Stowing means controlling the energy, placing the tool securely, organizing the hose and protecting others from a falling tool or protruding steel.

CP breaker main parts and letter key

Main parts



CP 1210



CP 1230 and 1240

A. Trigger (start/stop device)
B. Handle
C. Back head
D. Air intake
E. Muffler
F. Front head
G. Latch

Note. Atlas Copco Construction Tools / Chicago Pneumatic. (2010). CP 1210, CP 1230, CP 1240: Safety and operating instructions (9800 0608 90d, p. 12). Full source context.

Worked example · Count exposure honestly

Fictional log: 09:00–09:20 setup; 09:20–09:35 tool in use; 09:35–09:45 pause; 09:45–10:05 tool in use. The worker reports “65 minutes of hammer operation.”

Think it through.

Elapsed time is 65 minutes, while recorded tool-use intervals total 15 + 20 = 35 minutes. The remaining 30 minutes are setup/pause. Keep both values because different exposures may continue during non-trigger time. The log does not calculate a permissible vibration dose; actual assessment needs the relevant exposure data and method.

Try it and explain your thinking

Label the supply-to-tool diagram, including hose, coupling, retainer and tool

Label the supply-to-tool diagram, including hose, coupling, retainer and tool. Explain where stored energy remains after a trigger is released.

Skills you use: Reading, Problem solving.

Calculate and check the result

Recalculate the worked log and propose two columns that prevent elapsed-time and trigger-time confusion.

Skills you use: Numeracy, Writing.

Create a short illustrated handling/stowing brief

Create a short illustrated handling/stowing brief. A partner introduces a damaged coupling; revise the brief and save the corrected record.

Skills you use: Digital, Collaboration, Communication, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use the actual tool manual and assessed material, air supply and exposure controls.	0 / 1 / 2	Evidence/help: _____
Observe safe handling, PPE, people separation, controlled use and shutdown/stowing under qualified supervision.	0 / 1 / 2	Evidence/help: _____
Require the learner to identify a defect and stop; do not create a live hose failure for a demonstration.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B4, printed p. 30; source PDF pp. 32–32.
- Atlas Copco Construction Tools / Chicago Pneumatic. (February 2010; 9800 0608 90d). CP 1210, CP 1230 and CP 1240: Safety and Operating Instructions. Open publication.
- Chicago Pneumatic. (September 2005; 9800 0530 01). CP 1230 Demolition Tool: Spare Parts List. Open publication.
- Canadian Centre for Occupational Health and Safety. (retrieved October 30, 2026). Powered Hand Tools: Pneumatic Tools, Basic Safety. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 7: Noise, Vibration, Radiation and Temperature. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 12: Tools, Machinery and Equipment. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline B5

Safety working around and operating propane tanks

Propane equipment is an assembly: cylinder, valve, connector, regulator where required, hose and appliance must be compatible. A familiar-looking fitting is not enough to establish compatibility. Identify each part and the pressure/function it serves before preparing the task. The acquired Fairview Tiger Torch instructions and the actual appliance markings are useful examples; they do not authorize mixing arbitrary components.

- Potential fire hazards while working around propane tanks
- Proper storage and handling
- Proper connection procedures and check for leaks

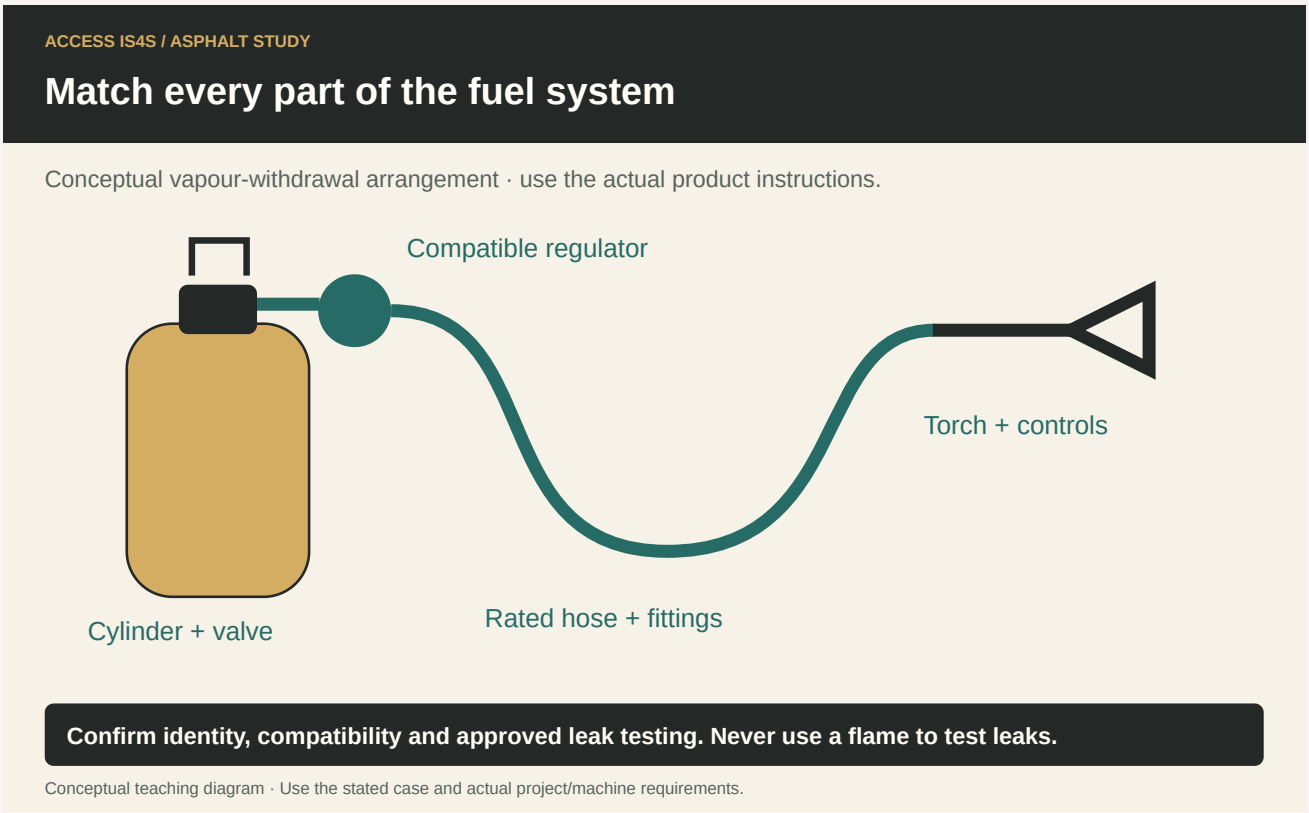
LEARNING STEP 1

Potential fire hazards while working around propane tanks

Propane is stored under pressure and can form a flammable mixture with air when released. Escaping liquid can cause cold injury; a leak can also create an oxygen-displacement concern in an unsuitable space. Ignition sources include flames, hot surfaces, sparks and some equipment operations. Adequate ventilation, separation and the correct assembly therefore belong in the task plan.

A cylinder must be suitable, identified and within the applicable inspection/requalification requirements. Inspect its condition and the connected equipment through the prescribed pre-use checks. Odour may indicate a release, but smell is not a complete leak or concentration test. An unexpected leak, damaged connection or uncertain assembly is a reason to stop and use the emergency/isolation procedure, not to experiment with ignition.

Match every part of the fuel system



Note. ACCESS IS4S. (2026). Match every part of the fuel system [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Proper storage and handling

Storage and transport are different contexts. Secure the cylinder against unintended movement in the required orientation, protect the valve and keep it in the permitted ventilated location away from prohibited heat/ignition exposures. Follow the supplier, workplace and transport requirements for the actual quantity and vehicle. The stowed torch and hose should be protected from damage; closing a valve alone does not make loose equipment adequately secured.

Identify whether the task involves use, temporary placement, storage or transport. Each can have different requirements. Keep the product SDS and relevant instruction accessible, and know the site's spill/fire/emergency arrangements. Do not place a cylinder in a vehicle space merely because it fits there.

LEARNING STEP 3

Proper connection procedures and check for leaks

Connection is a supervised procedural skill. Confirm the cylinder/appliance identity and the specified connector type, inspect sealing surfaces and the complete hose/regulator assembly, and use the manufacturer's connection method. Some fittings have different thread directions or sealing arrangements;

an improvised adapter, force or an unspecified sealant can create a hazard. The actual instructions determine the connection and tightening method.

Use the manufacturer-approved leak-test solution and test procedure; a flame is never a leak detector. Observe the specified connection locations and response to bubbles or another leak indication. If the assembly leaks, take it out of operation and follow the prescribed isolation/repair route. Rechecking after a repair is part of the applicable procedure. An instructor assesses the actual connection and leak check with approved equipment, not a learner's confidence alone.

Tiger Torch 95-B and 445 instruction sheet

TIGER TORCH

GAS-FLO by Fairview

Tiger Torch Ltd.
Operating Instruction and Safety Procedures
Tiger Torch Models 95-B and 445

CERTIFICATIONS

- CSA Approved Certificate #MC188901
- BTU rating 208,000
- 0-30 PSI operating pressure

ASSEMBLY INSTRUCTIONS TO PROPANE SOURCE

- The torch is equipped with a #53 orifice to operate using propane vapor fuel only.
- Not to be used on liquid feed propane source.
- Recommended proper assembly of the torch to the propane tank or source requires connecting the POL tank fittings to a 0-30 PSI regulator, GAS-FLO® Model GR-330. Connect the feed hose from the regulator to the torch valve. Hose and fittings that are ULC or CSA approved with maximum pressure rating of 350 PSIG should be used.
- Once assembled to tank, open tank valve and test assembly for leaks using a water spray.

OPERATION AND SAFETY PROCEDURES

- Ensure torch valve is closed and open valve on propane tank.
- Holding igniter in front of torch head, open torch valve until ignited.
- Open torch valve to desired heat source and adjust the regulator until torch burns with a clean blue-red flame.
- To shut down, close torch valve until flame is extinguished. When not in use, close propane tank valve.
- Torch should be held by handle grip only.
- Keep torch head away from hands or any flammable materials, as it is extremely hot during operation and for a period after shutdown. Do not lay torch on ice or snow when hot, as it may cause cracking of the head.
- Ensure torch head is adequately cooled prior to storage.
- If stored still connected to a propane source, ensure valve on torch and propane tank are both closed tightly
- Designed for outdoor use but can be used indoors if adequate air for combustion and ventilation is available.

Fairview www.Fairviewfittings.com

TIGER TORCH

GAS-FLO by Fairview

Tiger Torch Ltd.
Mode d'emploi et procédures de sécurité
Tiger Torch modèles 95-B et 445

CERTIFICATIONS

- CSA certifié - certificat #MC188901
- Débit thermique 208,000 BTU
- Pression de fonctionnement de 0-30 PSI

INSTRUCTIONS DE MONTAGE AVEC SOURCE DE GAZ PROPANE

- La torche est équipée d'un orifice #53 permettant l'usage de la torche avec les vapeurs de gaz propane seulement.
- Ne pas utiliser avec une source d'alimentation de propane liquide.
- Recommande' l'assemblage de la torche au gaz propane ou autre source nécessite les raccords du réservoir POL à un régulateur de 0 à 30 lb/po2, modèle GR-330 par GAS-FLO®. Connecter le tuyau d'alimentation du régulateur à la valve de la torche. Tuyaux et raccords doivent être certifiés ULC ou CSA s et indiquer un taux de pression maximale de 350 lb/po2 de résistance.
- Une fois assemblé au réservoir, ouvrez la soupape de propane et faire un test à l'aide d'un jet pour détecter s'il y a une fuite.

PROCÉDURES D'UTILISATION ET DE SÉCURITÉ

- Assurez-vous que la valve de la torche est fermée et ouvrir le robinet sur le réservoir de propane.
- Tenez l'allumeur en face de la tête du chalumeau, ouvrir le robinet de la torche jusqu'au moment de l'ignition de la flamme
- Ouvrir le robinet du chalumeau à la source de chaleur désirée et régler le gradateur jusqu'à ce que la flamme brûle avec une flamme propre bleu-rouge.
- Pour arrêter, fermez le robinet du chalumeau jusqu'à ce que la flamme s'éteigne. Fermez le robinet du réservoir de gaz propane lors que la torche n'est pas utilisée.
- En tout temps, assurez-vous de tenir la torche par la poignée seulement.
- Garder la tête de la torche éloignée des mains et de tout matériau inflammable, durant et après son utilisation. Ne pas mettre la tête de la torche sur la glace ou la neige lorsqu'elle est chaude, car cela peut causer des fissures à la tête.
- Assurez-vous que la tête de la torche est complêtement refroidie avant de la ranger.
- Si le réservoir est encore branché à une source de propane lorsque rangée, assurez-vous que les robinets de la torche et du réservoir de propane sont bien fermés
- Conçu pour une utilisation extérieure, mais peut être utilisé à l'intérieur à la condition que la ventilation et le débit d'air soient adéquats pour la combustion.

Fairview www.Fairviewfittings.com

Note. Tiger Torch Ltd. (n.d.). Operating instruction and safety procedures: Models 95-B and 445 [Bilingual instruction sheet, Fairview-hosted]. Full source context.

Worked example · The assembly register

Fictional teaching sample: the cylinder and torch are identified, but the hose rating is unreadable and the regulator model cannot be matched to the appliance. A learner proposes connecting them because the threads fit.

Think it through.

The assembly has two unresolved compatibility/condition questions. Matching threads does not establish pressure suitability or the correct sealing design. Keep the assembly out of operation; obtain the appropriate component identification and manufacturer instructions, then have the supervised setup and leak test completed. Record the unknown components rather than replacing unknown values with guesses.

Try it and explain your thinking

Trace the labelled propane assembly and state the function of each part

Trace the labelled propane assembly and state the function of each part. Identify which parts must be matched to the actual appliance.

Skills you use: Reading, Communication.

Write a connection-and-leak-check preparation record for the worked case

Write a connection-and-leak-check preparation record for the worked case. State the rejection conditions and the records required before use.

Skills you use: Writing, Problem solving.

Inspect the actual retained torch instructions on a device and save the publication/page identity

Inspect the actual retained torch instructions on a device and save the publication/page identity. Explain to a partner how use, storage and transport questions differ; improve the record after their feedback.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Instructor confirms compatible cylinder, appliance, connections, condition, location, fire controls and emergency arrangements.	0 / 1 / 2	Evidence/help: _____
Observe the approved connection, leak test, response to a simulated defect, isolation and secure stowing/transport preparation.	0 / 1 / 2	Evidence/help: _____
Record the actual equipment, procedure, support and outcome; no live leak is deliberately created.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B5, printed p. 31; source PDF pp. 33–33.
- Tiger Torch Ltd.. (n.d.; Fairview-hosted sheet). Operating Instruction and Safety Procedures: Models 95-B and 445. Open publication.
- BC Forest Safety Council. (n.d.; publisher 2022 hosting path). Safe Use of Tiger Torches. Open publication.
- Canadian Centre for Occupational Health and Safety. (retrieved October 30, 2026). Chemical Profiles: Propane. Open publication.

- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 5: Chemical and Biological Substances. Open publication.
- Transport Canada. (Web snapshot: October 30, 2026). Transportation of Dangerous Goods: Training. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline B6

Weather effects and precautions

Weather affects the worker as well as the material. Hot mix and hot pavement add radiant heat to the work environment; sun, humidity, air movement, workload and clothing affect how the body loses heat. A forecast temperature alone cannot describe the exposure. Crew members also differ in acclimatization and health needs, which should be supported without demanding personal medical details in front of the group.

- The effects of temperature on the body
- The importance of proper hydration
- Define effects of exposure to direct sun

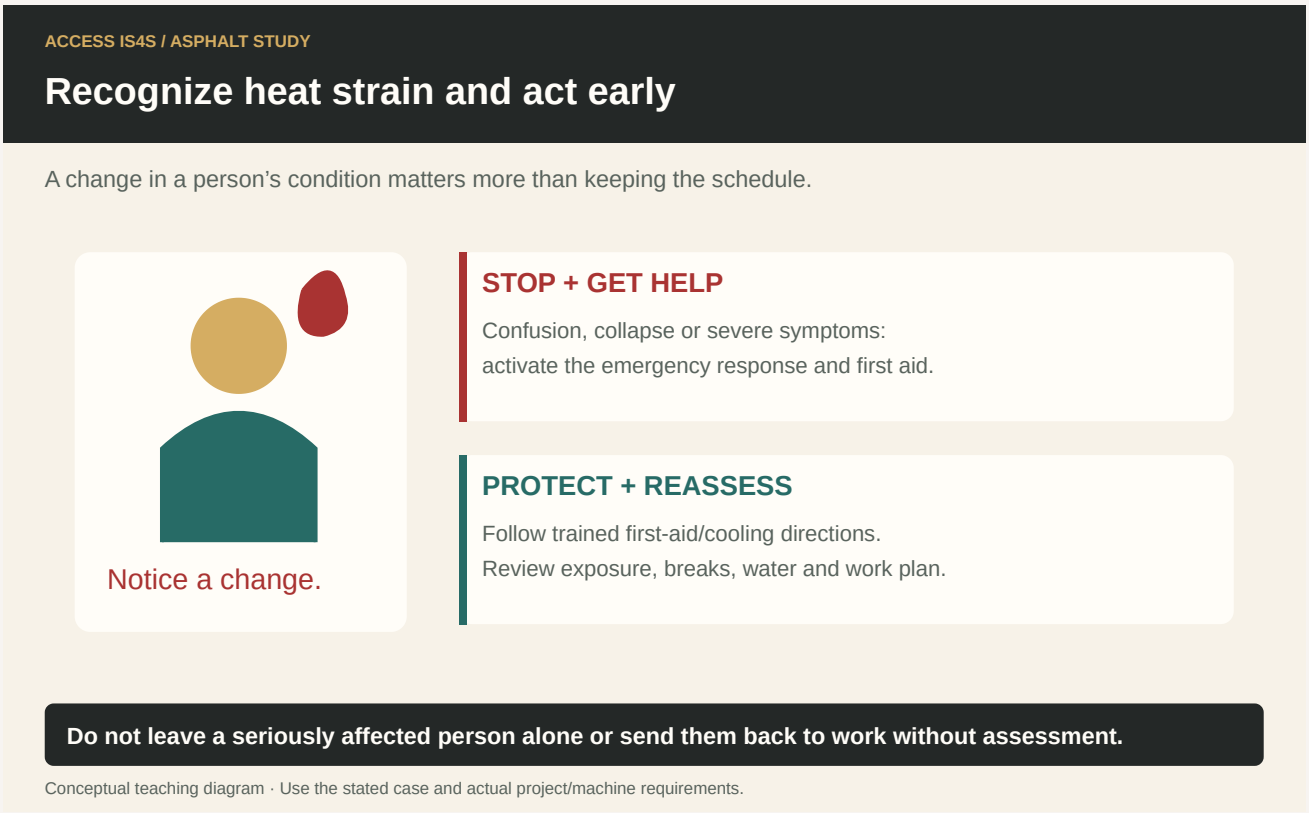
LEARNING STEP 1

The effects of temperature on the body

The body produces heat during work and loses heat through exchanges with the surroundings, including evaporation of sweat. Heavy work, radiant heat and clothing can make cooling more difficult. Heat strain can develop even when the air temperature seems familiar. The work plan assesses conditions, provides appropriate controls and adjusts work/rest and supervision arrangements as conditions change.

Recognize changes in behaviour, weakness, dizziness, headache, nausea or collapse and obtain help. Cold, rain and wind can also affect dexterity, judgement, traction and health. A weather response includes clothing, shelter, suitable work organization and an emergency plan. A new worker needs an acclimatization arrangement rather than an assumption that determination will make them tolerate the exposure.

Recognize heat strain and act early



Note. ACCESS IS4S. (2026). Recognize heat strain and act early [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

The importance of proper hydration

Workers need ready access to appropriate drinking water and the opportunity to drink regularly under the workplace heat plan. Hydration is one part of control; it does not replace reducing exposure, rest, cooling or recognizing illness. Do not make a class-wide medical fluid prescription or assume a person with symptoms only needs another drink.

Heat exhaustion and heat stroke require prompt response. The WorkSafeBC heat-stress guide treats heat stroke as a medical emergency. Altered mental state, collapse or other severe signs require emergency assistance: call the local emergency number, get the first-aid attendant and begin the plan's safe cooling response while waiting for help. Do not leave the person alone or give fluids to someone unable to swallow safely. The trained response follows the workplace emergency and first-aid arrangements.

LEARNING STEP 3

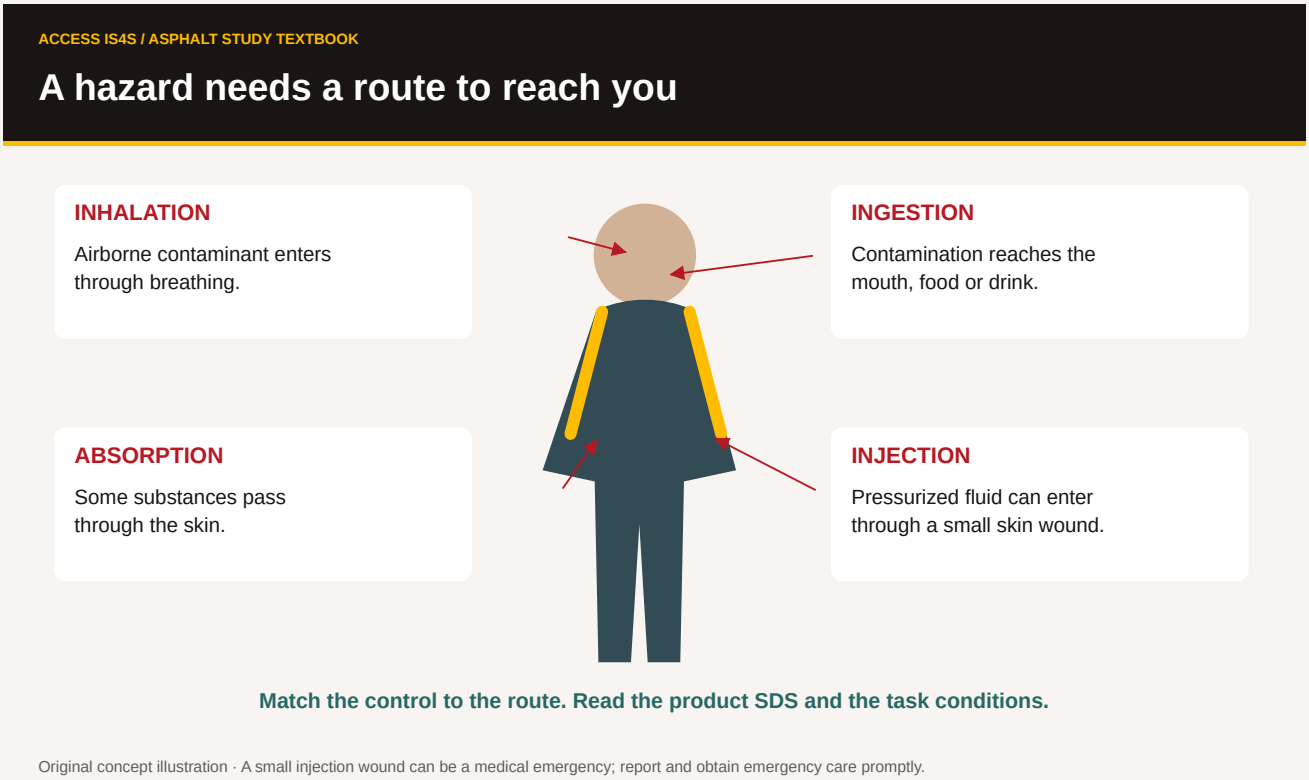
Define effects of exposure to direct sun

Ultraviolet exposure can damage skin and eyes, including on days that do not feel extremely hot. The task plan combines suitable clothing/head protection, shade, scheduling and appropriate sunscreen or other protection under the guidance used by the workplace. Select protective items so they remain compatible

with required PPE and the task.

A reflective or hot work surface adds to the need to assess exposure. Check breaks, water, shelter and communication before starting; recheck when work intensity, weather or worker condition changes. A short team check-in should invite early reporting without blame. Care for the crew includes making it possible to use the controls, not only telling people they exist.

A hazard needs a route to reach you



Note. ACCESS IS4S. (2026). A hazard needs a route to reach you [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Worked example · A break plan that cannot be used

Fictional schedule: a crew is assigned continuous handwork from 10:00 to 12:00. The written heat plan requires planned cool-rest opportunities, but the relief person has been reassigned. A new worker begins making confused replies.

Think it through.

The plan is not being implemented, and confusion is an emergency warning sign in this context. Stop the affected work, summon emergency/first-aid help and follow the cooling response. Arrange coverage and reassess the work before resuming. The two-hour scheduled block does not establish a safe exposure interval; it reveals a mismatch between the plan and staffing.

Try it and explain your thinking

Explain and apply the idea

List five factors beyond air temperature that influence heat risk, then match each to information the supervisor must consider.

Skills you use: Reading, Problem solving.

Mark the time span in the worked case and identify why the plan failed

Mark the time span in the worked case and identify why the plan failed. Write the first emergency communication with location and observed condition.

Skills you use: Numeracy, Writing, Communication.

Develop a weather check-in card with heat, sun and cold/wet prompts

Develop a weather check-in card with heat, sun and cold/wet prompts. Test it with a partner, revise it and save a version that leaves medical information private.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Confirm the current heat/cold exposure and emergency plans before outdoor practice.	0 / 1 / 2	Evidence/help: _____
Observe recognition and reporting in a spoken scenario, access to water/rest/shade and safe use of selected protection.	0 / 1 / 2	Evidence/help: _____
Qualified first-aid instruction assesses treatment skills; this chapter assesses recognition, prevention and prompt escalation.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B6, printed p. 32; source PDF pp. 34–34.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 7: Noise, Vibration, Radiation and Temperature. Open publication.
- WorkSafeBC. (2026; R05/26). Preventing Heat Stress at Work. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline B7

Safety and traffic control procedures

Traffic control is a planned system for road users, workers and equipment. Signs, devices, traffic control persons (TCPs), traffic assistants and communication have defined roles. A paving learner must understand that system even when they are not assigned to control traffic. At a community entrance, include pedestrians, cyclists, mobility needs and emergency access in the planning questions.

Traffic control procedures (TCP) Certification

Review TCP training content

Review TCP regulations

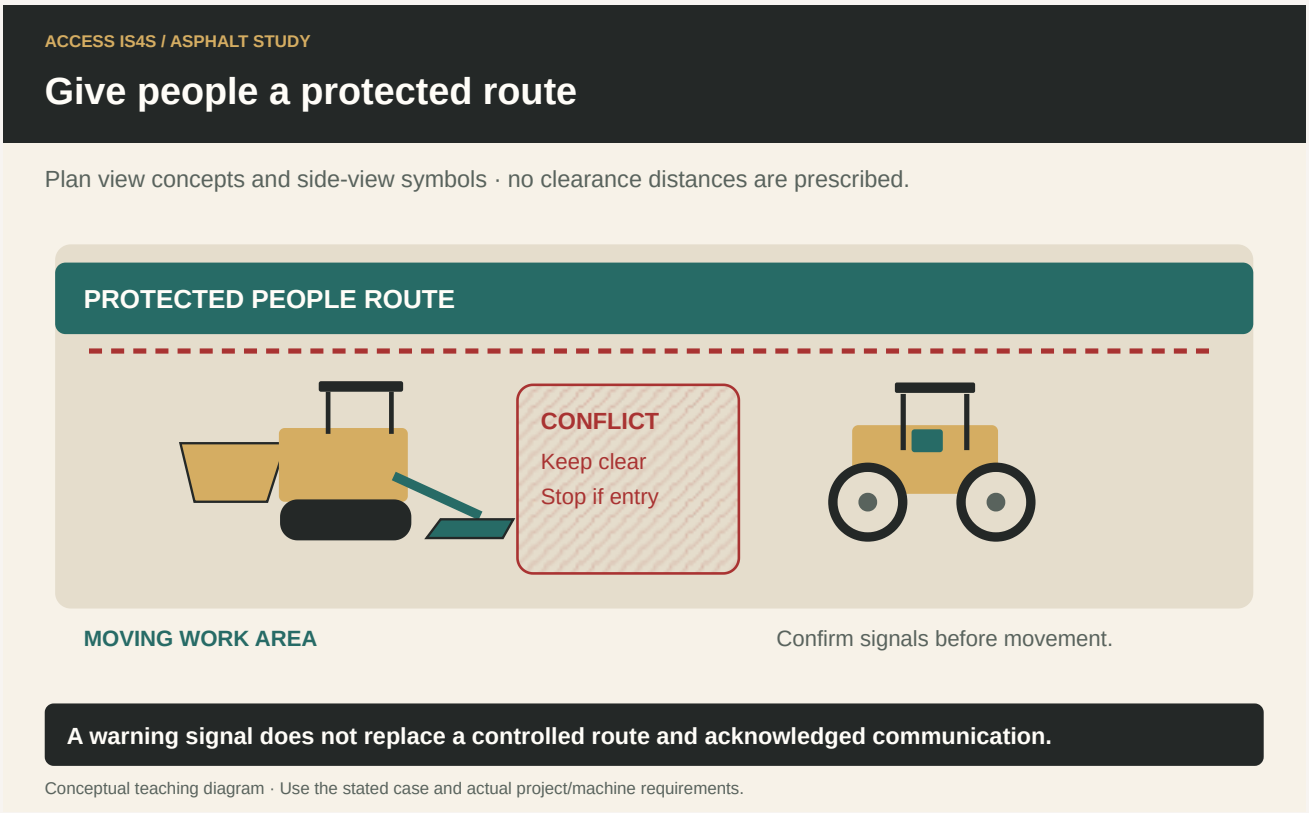
LEARNING STEP 1

Traffic control procedures (TCP) Certification

WorkSafeBC Part 18 defines traffic-control training and responsibilities. Section 18.6.2 addresses a traffic-control-person training program, including knowledge and practical competency assessment; the employer must meet the applicable qualification requirements. A classroom paving chapter does not issue that qualification. Confirm the accepted training program and the actual person's current evidence before assigning TCP duties.

A traffic assistant has a different role and workplace training requirements under 18.6.3. Do not relabel an unqualified person a TCP because the crew needs help. Course completion records, employer orientation and site-specific instruction serve different purposes; all applicable requirements must be met.

Give people a protected route



Note. ACCESS IS4S. (2026). Give people a protected route [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Review TCP training content

Traffic-control training develops recognition of hazards, the required plan, setup and removal of devices, safe positioning and escape routes, signs/signals, communication and responses to changing conditions. The B.C. Traffic Management Manual for Work on Roadways provides the roadwork framework, used with current regulation and road-authority requirements.

The learner reads the approved plan's location, revision, work activity and traffic arrangement before interpreting an individual sign. A diagram from a manual is a typical arrangement with conditions, not a ready-made plan for every site. Conditions such as sight distance, road speed, intersections, queues, pedestrians and lighting affect the actual plan.

LEARNING STEP 3

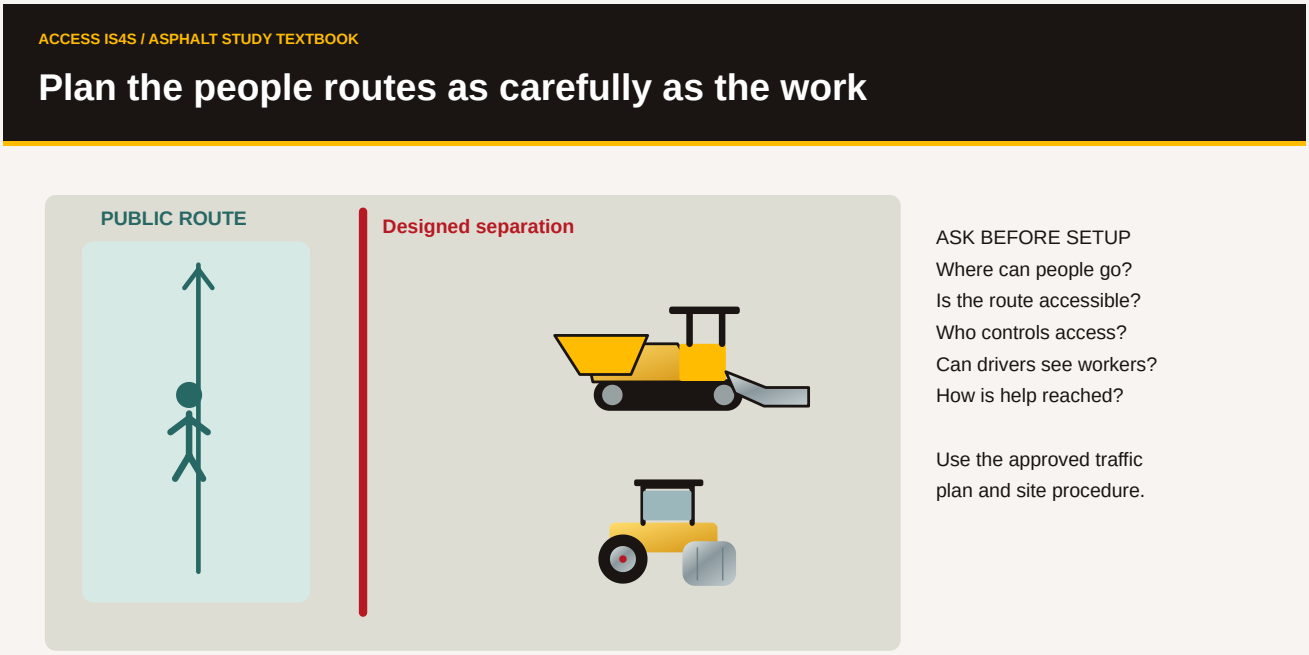
Review TCP regulations

The number and positioning of people and devices follow the risk assessment, applicable requirements and approved traffic-control arrangement. There is no universal answer of one or two TCPs for all paving work. Under 18.8, the position must be identified as safe in the risk assessment; escape and approach visibility

matter. Equipment is kept clean and serviceable. Precise, deliberate signals and effective communication between multiple TCPs are required by the relevant provisions.

The paver crew must not change the traffic arrangement informally when production slows or a truck arrives. Communicate the changing need to the responsible person, maintain the controls and wait for an authorized adjustment. A queue extending beyond the planned warning arrangement is a changed condition requiring action, not a reason to hurry signals.

Plan the people routes as carefully as the work



Note. ACCESS IS4S. (2026). Plan the people routes as carefully as the work [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Worked example · The plan revision matters

Fictional documents: T-02 revision A places the facility entrance outside the work area. Revision B extends the work beyond that entrance. The crew has B, but a traffic briefing still uses A.

Think it through.

The drawings describe different access conditions. Before work under the changed arrangement, resolve the mismatch through the responsible traffic-control planner/supervisor and brief all affected roles. Record the current approved revision and the entrance provisions. An extra cone added by an unassigned person does not resolve the plan conflict.

Try it and explain your thinking

Explain and apply the idea

Compare TCP and traffic-assistant training requirements using retained Part 18 sections 18.6.2–18.6.3. Identify what qualification this chapter does not issue.

Skills you use: Reading, Writing.

Calculate and check the result

Annotate the work-zone concept with questions about warning, transition, work area, public route and return to normal traffic. Explain why no spacing can be taken from it.

Skills you use: Problem solving, Communication.

Save a version register for the worked case, then role-play a revision handover

Save a version register for the worked case, then role-play a revision handover. Ask the listener to find the current entrance instruction and improve the brief if they cannot.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Check the actual site plan and role qualifications before road-side observation.	0 / 1 / 2	Evidence/help: _____
Observe correct interpretation, staying within assigned duties and communication of a changed condition.	0 / 1 / 2	Evidence/help: _____
Where TCP competence is assessed, use the approved training/assessment program and qualified assessor, not this classroom record alone.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B7, printed p. 33; source PDF pp. 35–35.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 18: Traffic Control. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Guidelines, Part 18. Open publication.
- BC Ministry of Transportation and Infrastructure. (2020). Traffic Management Manual for Work on Roadways. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Traffic Management Manual Updates. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline B8

Plant site and gravel pit safety

A plant or gravel pit is an interconnected worksite. Material moves through feeders, crushers, screens, conveyors, stockpiles and vehicles; people may not be able to see the next process from where they stand. A stoppage at one machine can leave others running or material stored above it. Begin with the site induction, responsible supervisor and applicable jurisdiction. A mine/pit operation may be governed by B.C.'s mining safety framework; other workplaces fall under the applicable WorkSafeBC requirements.

Define noise hazards and precautions

Site specific hazards

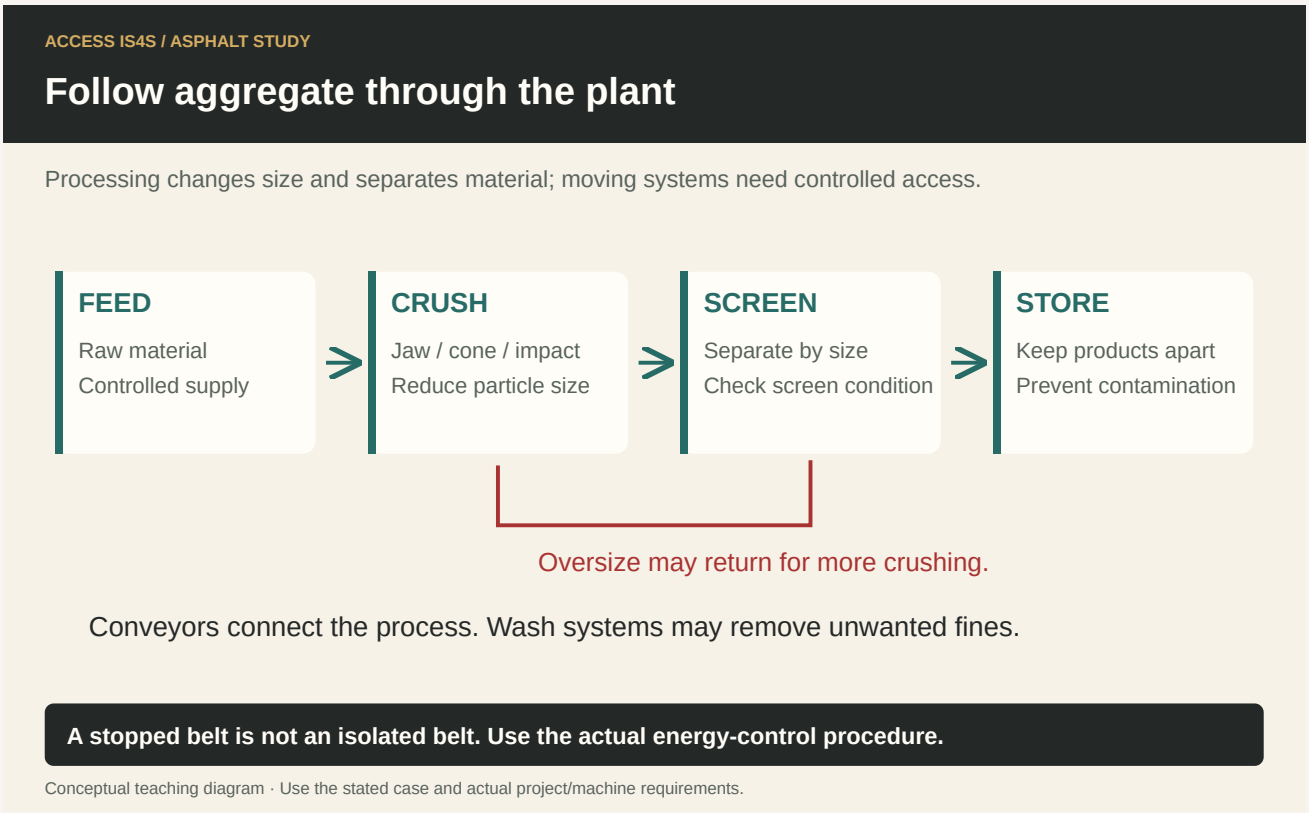
LEARNING STEP 1

Define noise hazards and precautions

Crushing, screening, engines and material impacts can create sustained or changing noise. Hearing protection is selected within the site’s assessed program. It must remain compatible with communication and other PPE; a person who cannot hear a warning needs a safe communication arrangement rather than removing protection in the exposure area.

Review the site’s noise areas, permitted access, signs and protection requirements before entering. Equipment condition and engineering controls influence noise as well as reliability. An unusual sound can be a defect indicator, but approaching the source through a restricted area to investigate is not an appropriate response for an unassigned learner.

Follow aggregate through the plant



Note. ACCESS IS4S. (2026). Follow aggregate through the plant [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

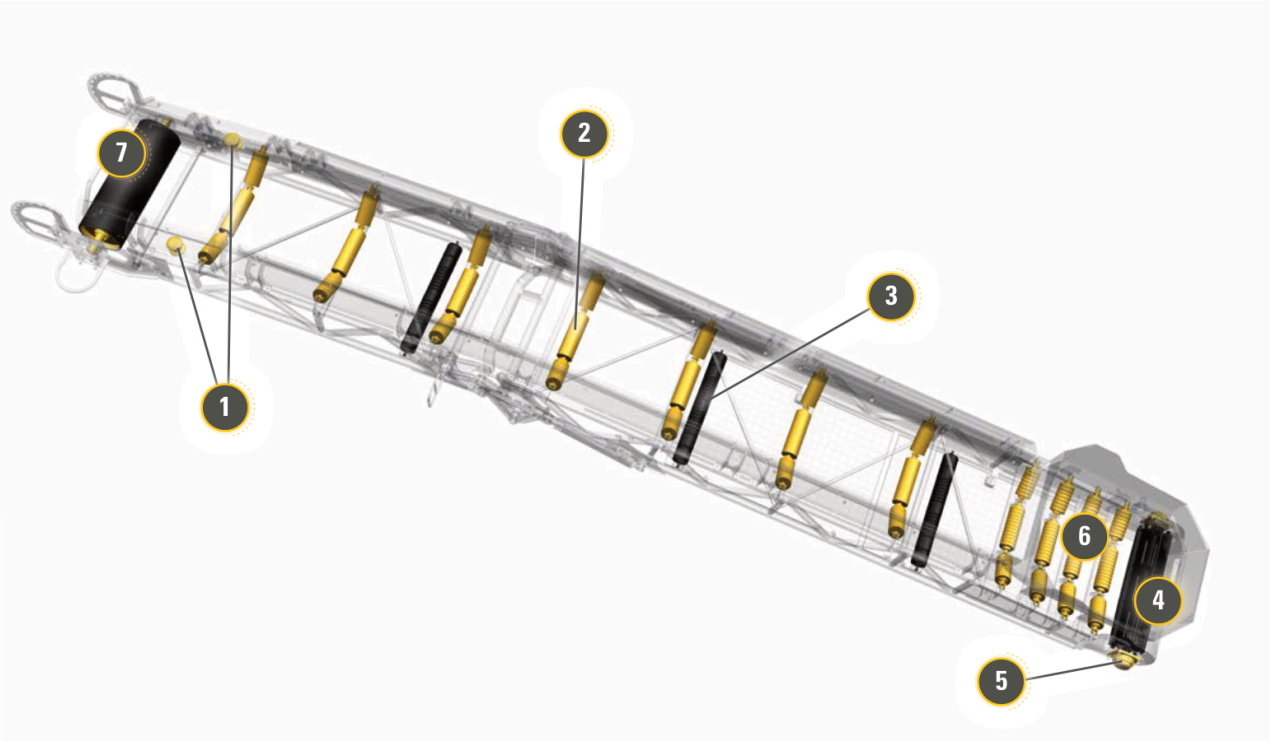
LEARNING STEP 2

Site specific hazards

Crushing and entanglement hazards occur at feed openings, belt/pulley nip points, rotating shafts and moving mechanisms. Guards do not authorize reaching through or around them. Isolation must account for connected equipment and stored material/energy when cleaning or maintenance is assigned. A bridge of material can collapse; an idle belt may still be connected to a starting sequence.

Vehicle/loader routes, stockpile faces and reversing areas require the site's controls. Flying debris and dust can extend beyond a machine footprint; assess the material, collection/suppression systems and exposure. Stockpile instability, water edges and restricted plant structures also matter. Follow approved pedestrian routes and do not climb a pile or conveyor to take a better look. Report missing guards, damaged dust controls or access conflicts to the responsible person without entering the hazard.

Every numbered component in this conveyor diagram



Note. Caterpillar. (2026). Cold Planer Parts Reference Guide (QEXQ3131-01 (05/2026), source PDF p. 25). Faithful crop; original labels retained. PM310, PM312, PM313, PM620, PM622, PM820, PM822 and PM825; individual parts/model tables apply. Full source context.

Worked example · One stopped belt is not the whole system

Fictional process: feeder → crusher → conveyor → screen → stockpile. The conveyor stops and material begins accumulating upstream. A visitor proposes stepping over it while an operator investigates.

Think it through.

A stopped component does not establish isolation or a safe crossing. Accumulation adds a stored-material hazard and connected equipment may restart. Keep the visitor on the authorized route, report the condition and use the site's controlled shutdown/isolation response. The process map helps identify connected sources; it does not replace the isolation plan.

Try it and explain your thinking

Explain and apply the idea

Trace the process diagram and identify four different hazard mechanisms, including one that remains when motion stops.

Skills you use: Reading, Problem solving.

Write a visitor brief for the worked site using only authorized routes and role boundaries

Write a visitor brief for the worked site using only authorized routes and role boundaries. Include the response to an unfamiliar alarm.

Skills you use: Writing, Communication.

Create a digital process-and-access sketch

Create a digital process-and-access sketch. A partner removes a guard symbol and blocks a pedestrian route. Record the defects and revise the access plan through the responsible role.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Confirm jurisdiction, induction, PPE and permitted observation area for the actual plant/pit.	0 / 1 / 2	Evidence/help: _____
Observe route compliance, recognition of noise/dust/mobile-equipment/connected-machine hazards and appropriate reporting.	0 / 1 / 2	Evidence/help: _____
Maintenance or confined-space activities require their own authorized instruction and assessment.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B8, printed p. 34; source PDF pp. 36–36.
- BC Ministry of Mining and Critical Minerals. (June 2026). Health, Safety and Reclamation Code for Mines in British Columbia. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 12: Tools, Machinery and Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- National Institute for Occupational Safety and Health. (2015; Publication 2015-105). Best Practice Engineering Control Guidelines to Control Worker Exposure to Respirable Crystalline Silica during Asphalt Pavement Milling. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline B9

Light vehicle transporting procedures

Transport preparation begins with a complete description: vehicle, trailer, load, mass, dimensions, attachments, route and driver. **Gross vehicle weight rating**, **actual loaded mass**, **axle rating**, **tow rating** and **cargo-securement rating** answer different questions. A small-looking load can exceed one component's limit even when the whole combination appears capable.

The 5,000 kg categories used in the learning tasks are a comparison exercise, not a single legal switch between company and federal rules. Applicable B.C. vehicle, driver, inspection, cargo-securement and permit requirements depend on the actual combination and activity. Company rules apply alongside the law and cannot reduce its requirements.

Procedures for transportation of vehicles under 5000 kg

Procedures for transportation of vehicles over 5000 kg

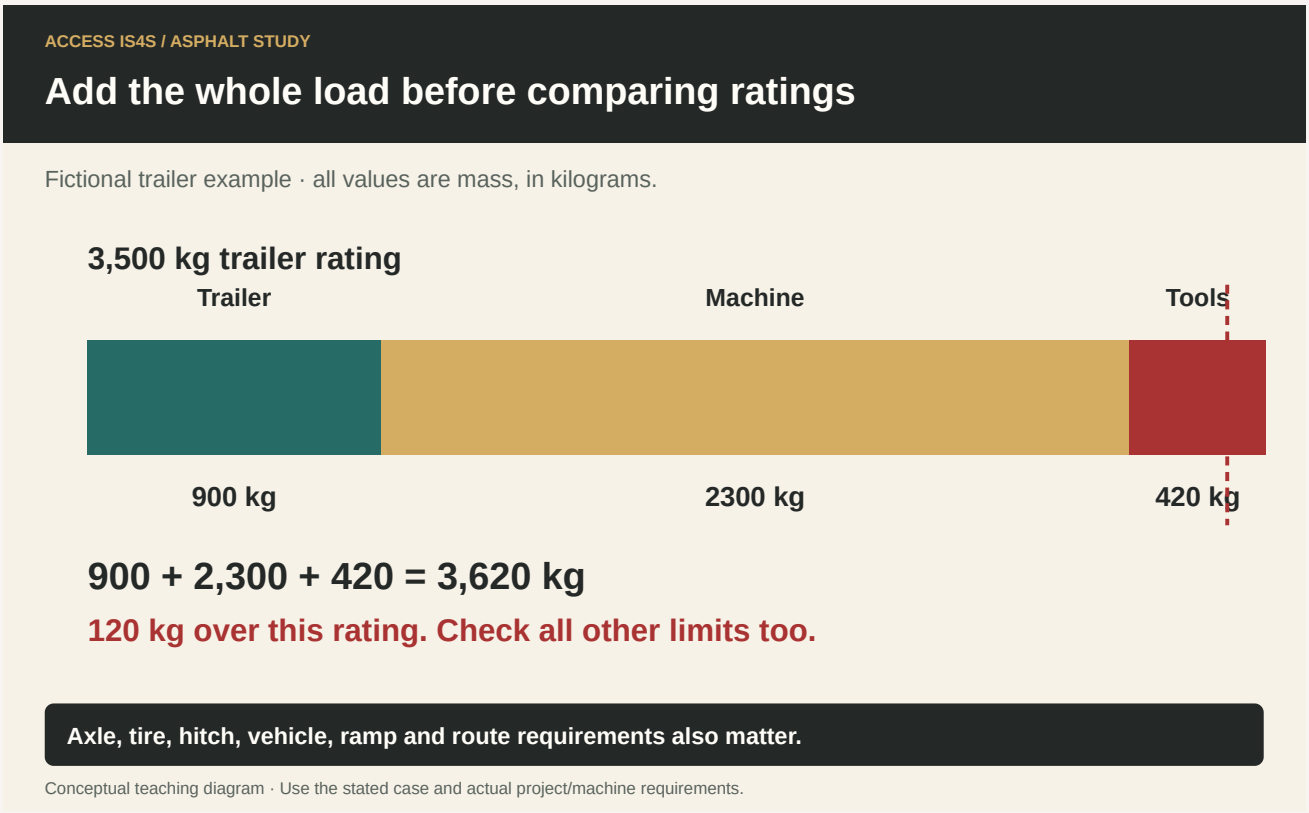
LEARNING STEP 1

Procedures for transportation of vehicles under 5000 kg

For a light vehicle or combination, identify the manufacturer's ratings, registration/licensing requirements, trailer/hitch/brake/tyre/axle suitability and the actual loaded condition. Include people, fuel, tools and attachments in the appropriate mass assessment. A nominal machine mass may exclude options or carried material. Verify the route and permitted load dimensions before departure.

Cargo must be restrained by an arrangement meeting applicable requirements and the actual anchors and tiedowns. Keep loose tools, ramps and attachments secure too. A company checklist helps organize the checks, but a tick beside 'under 5,000 kg' cannot establish that the combination is legal or safe. The driver and responsible loading personnel need a common load record.

Add the whole load before comparing ratings



Note. ACCESS IS4S. (2026). Add the whole load before comparing ratings [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

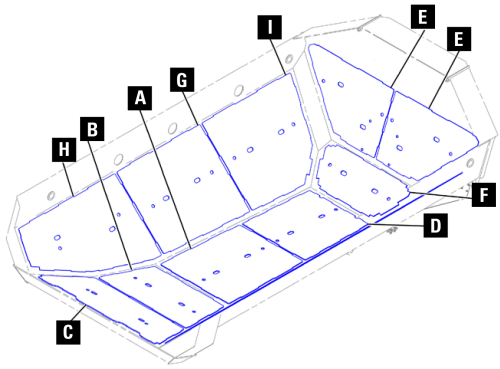
Procedures for transportation of vehicles over 5000 kg

Heavier transport adds capacity and planning questions but does not erase the same fundamentals. Determine the relevant vehicle and cargo-securement rules, driver qualifications, required inspections and any oversize/overweight permits or route restrictions. The B.C. cargo-securement information and National Safety Code Standard 10 provide the applicable framework; special commodity/equipment provisions may matter.

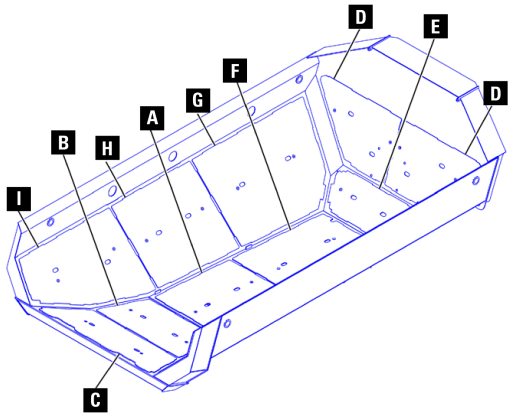
The weakest relevant component or a prohibited configuration can control the outcome. A tiedown's labelled capacity is not automatically the allowable restraint of the whole system: angle, method, anchor and applicable rules matter. Loading/unloading is also a separate controlled operation, with suitable ground, ramps, alignment, separation and assigned communication. The trip record must reflect the actual loaded machine, not a previous similar delivery.

Articulated truck: dump-body liner locations

DUMP BODY LINERS



DUMP BODY LINERS



DUMP BODY LINERS

Sales Model	Serial Number Range	Item Location	Part Number	Quantity
740 GC	3T800001-05400 3F800001-05100	A	143-4002	1
		B	233-6950	1
		C	233-6951	1
		D	301-4831	1
		E	398-9914	2
		F	398-9915	1
		G	435-5703	2
		H	435-5709	2
		I	545-5274	2

DUMP BODY LINERS

Sales Model	Serial Number Range	Item Location	Part Number	Quantity
745	3T600001-05600 3F600001-05500	A	143-4002	1
		B	233-6950	1
		C	233-6951	1
		D	398-9914	2
		E	398-9915	1
		F	398-9917	1
		G	398-9918	2
		H	435-5703	2
		I	435-5709	2

Note. Caterpillar. (2022). Articulated Trucks — Parts Reference Guide (None, source PDF p. 9). Faithful crop; original labels retained. 725, 730, 730 EJ, 735, 740 GC, 740 EJ, 745; listed 3T/3F serial prefixes. Full source context.

Worked example · The trailer rating check

Fictional ratings-only exercise: trailer permitted gross mass 3,500 kg; measured empty trailer 900 kg; proposed machine 2,300 kg; tools/ramps 420 kg. Ignore no other real requirement; this is one arithmetic screening check.

Think it through.

Proposed gross mass = 900 + 2,300 + 420 = 3,620 kg. It exceeds the stated trailer figure by 120 kg, so the proposed loading fails this check. The unused-looking deck space does not change the result. Even a revised load below 3,500 kg would still require axle, tow, hitch, restraint, route and driver checks.

Try it and explain your thinking

Explain why “under 5,000 kg, therefore only company rules apply” is an inadequate transport decision

Explain why “under 5,000 kg, therefore only company rules apply” is an inadequate transport decision. Make a document checklist for either size category.

Skills you use: Reading, Writing, Problem solving.

Rework the example if tools/ramps are reduced to 260 kg

Rework the example if tools/ramps are reduced to 260 kg. State precisely what has and has not passed.

Skills you use: Numeracy, Communication.

Build a versioned load record, ask a partner to add a new attachment and recalculate

Build a versioned load record, ask a partner to add a new attachment and recalculate. Save/reopen the revision and explain who must approve the transport arrangement.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use actual ratings, measured/verified load information, suitable vehicle/trailer and the applicable transport plan.	0 / 1 / 2	Evidence/help: _____
Observe pre-trip/load checks, securement inspection and communication within the learner's authorized role.	0 / 1 / 2	Evidence/help: _____
Driver qualification and operating competence are assessed under the relevant vehicle requirements.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B9, printed p. 35; source PDF pp. 37–37.
- Province of British Columbia. (Web snapshot: October 30, 2026). Cargo Securement. Open publication.
- Canadian Council of Motor Transport Administrators. (June 2013). National Safety Code Standard 10: Cargo Securement. Open publication.
- Canadian Council of Motor Transport Administrators. (May 2016). Cargo Securement Standard: Interpretations and Guidance. Open publication.
- Transport Canada. (Web snapshot: October 30, 2026). Transportation of Dangerous Goods: Training. Open publication.

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Study chapter · Outline B10

Complete fatigue awareness training

Fatigue is reduced physical or mental capacity associated with inadequate sleep, prolonged wakefulness, demanding work or other factors. It can affect attention, reaction, memory and judgement before a person recognizes the full change. A crew depends on accurate handovers and timely responses; fatigue therefore belongs in work planning, not in a judgement about someone's character.

- Define mental fatigue
- Define a work schedule that helps reduce fatigue
- Discuss available training

LEARNING STEP 1

Define mental fatigue

Sleep debt accumulates when sleep is less than a person's need. **Circadian rhythms** influence alertness across the day and night. A worker may have a scheduled rest interval but little opportunity to sleep after commuting, meals and personal responsibilities. Extended wakefulness and successive demanding shifts can compound the problem.

Signs may include slowed responses, difficulty concentrating, errors, irritability or difficulty staying awake. These observations do not diagnose the cause. Respond through the workplace fitness-for-task and support arrangements, with privacy and respect. Caffeine or willpower cannot be treated as a complete replacement for sleep. A fatigued person may also be unsafe to drive home; plan a safe response rather than simply sending them onto the road.

Time off is not all sleep time



Note. ACCESS IS4S. (2026). Time off is not all sleep time [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Define a work schedule that helps reduce fatigue

A fatigue-aware schedule considers shift length, timing, consecutive duties, recovery opportunity, travel and workload. The employer's planning and response arrangements should make reporting possible and provide a practical way to change the assignment when needed. A schedule can look adequate on paper while overtime, an early call-out or a long commute removes the intended recovery.

Distinguish time off duty from sleep opportunity and actual sleep. A calculation can expose an obvious planning conflict, but it cannot certify that an individual is fit. Ask whether the schedule remains workable after a delay and whether the highest-consequence tasks have appropriate staffing and supervision. Record the changed assignment and handover so the response does not create another gap.

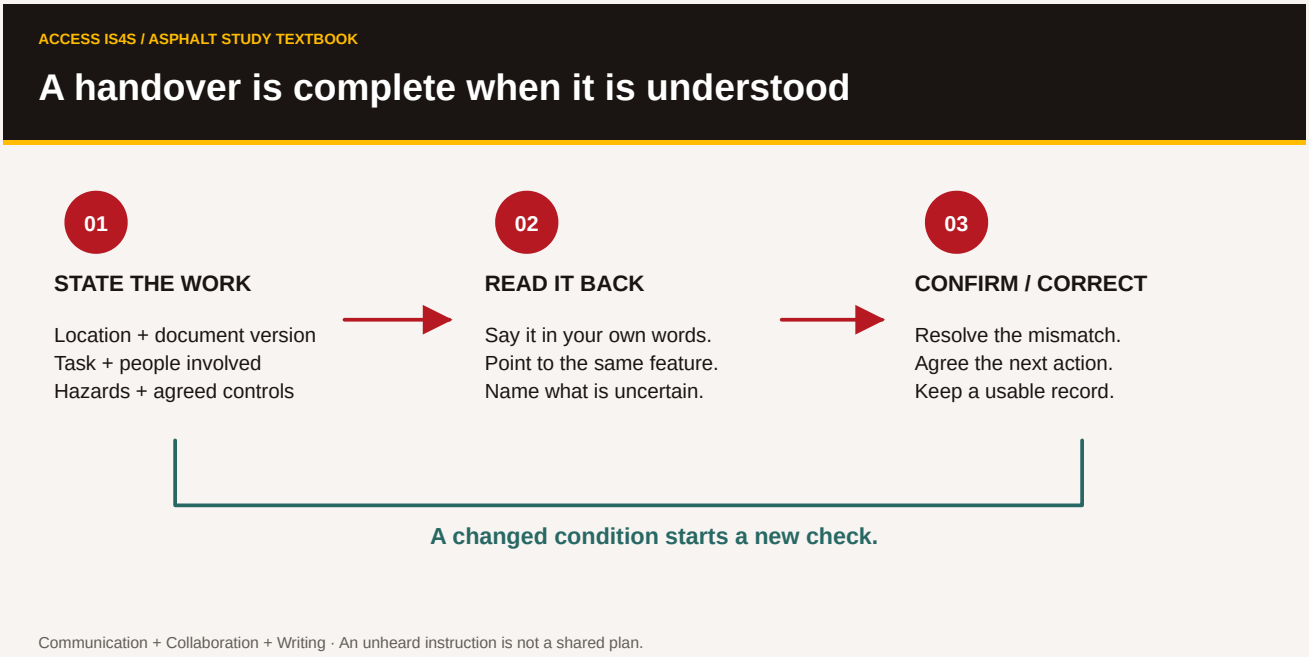
LEARNING STEP 3

Discuss available training

Useful fatigue-awareness training explains causes, signs, work and driving effects, reporting, prevention and the workplace response. CCOHS provides a retained evidence source for this study. Evaluate an external course by its content, current evidence, instructor/provider suitability and relevance to the actual work; a website name or completion badge alone does not establish effective controls.

The learner practises recognizing a condition, communicating it without blame and using the agreed support route. The supervisor practises responding to the work risk rather than demanding private information in front of colleagues. A training record identifies the content completed, the learner's demonstrated understanding and the employer-specific arrangements still required.

A handover is complete when it is understood



Note. ACCESS IS4S. (2026). A handover is complete when it is understood [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Worked example · Rest interval versus sleep opportunity

Fictional shift finishes at 20:00; the next begins at 06:00. Total travel uses 2 hours, meals/personal preparation 1.5 hours and a required call 0.5 hour. Calculate the maximum remaining interval before other interruptions.

Think it through.

There are 10 hours between shifts. Subtracting 2 + 1.5 + 0.5 leaves at most 6 hours available. That is an opportunity estimate, not proof of six hours of sleep or a universal fitness threshold. The schedule needs review through the workplace fatigue process, especially before safety-critical work or driving.

Try it and explain your thinking

Explain sleep debt and circadian timing in your own words

Explain sleep debt and circadian timing in your own words. Separate observed signs from a diagnosis or a judgement about a person.

Skills you use: Reading, Communication.

Recalculate the worked case if a late task adds 90 minutes

Recalculate the worked case if a late task adds 90 minutes. Explain the scheduling implication without declaring an individual fit or unfit from arithmetic alone.

Skills you use: Numeracy, Writing, Problem solving.

Compare the retained CCOHS content with your workplace training checklist

Compare the retained CCOHS content with your workplace training checklist. Create, peer-test and save a fatigue-reporting flow that includes a safe transport-home question.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use fictional scenarios for classroom privacy; explain the actual confidential reporting route before workplace practice.	0 / 1 / 2	Evidence/help: _____
Observe recognition, respectful communication, schedule checking and a safe handover/reassignment response.	0 / 1 / 2	Evidence/help: _____
Record completed awareness learning separately from a medical judgement or a vehicle-driving qualification.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, B10, printed p. 36; source PDF pp. 38–38.
- Canadian Centre for Occupational Health and Safety. (retrieved October 30, 2026). Fatigue. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 4: General Conditions. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Integrated practice · all nine Skills for Success

Prepare a clear work-zone briefing

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional community-access project has a 90 m × 4 m work strip. The plan protects a walking route, uses a designated truck guide and requires a working broom dust-control system. At 08:20, the guide's radio fails and the broom dust-control indicator shows a fault. A draft note says "continue while the guide shouts." The crew's shift gap is 11 hours, with 2 hours travel and 2 hours other responsibilities. No safe clearance or operating temperature is supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Change card · Use after the first attempt

The protected walking route must move because an entrance is needed; the responsible lead has not yet approved a new route.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person's actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.